distribution we have to see the properties of probability distribution. This section focuses on how to calculate these summary measures. These measures can be calculated using

(i) Mathematical Expectation and variance.

(ii) Moment Generating Function .

(iii) Characteristic Function.

9.6.1 Expectation of Discrete random variable

Let X be a discrete random variable which takes the values $x_1, x_2 \ldots\ldots x_n$ with respective probabilities $p_1, p_2 \ldots\ldots p_n$ then mathematical expectation of X denoted by $E(X)$ is defined by

$$E(X) = x_1 p_1 + x_2 p_2 + \ldots + x_n p_n$$

$$= \sum_{i=1}^{n} x_i p_i \qquad \text{where } \sum p_i = 1$$

Sometimes $E(X)$ is known as the mean of the random vairable X.

Result:

If $g(X)$ is a function of the random variable X, then $E\ g(X) = \sum g(x)p(X = x)$

Properties:

(i) $E(c) = c$ where c is a constant

Proof :

$$E(X) = \sum x_i p_i$$

$$E(c) = \sum c p_i = c \sum p_i = c \times 1 = c$$

(ii) $E(cX) = cE(X)$, where c is a constant

Proof:

$$E(X) = \sum x_i p_i$$

$$E(X) = \sum c x_i p_i$$

$$= c \sum x_i p_i$$

$$= c\ E(X).$$

(iii) $E(aX+b) = aE(X)+b$